R Programming Practical Examination

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2026-11-11

Part A: Basic Operations

A1 Create v as the sequence 5, 10, 15, ..., 50 using seq(). Print its length, sum and mean.

```
v <- c(5, 10, 15, 20, 25, 30, 35, 40, 45, 50)
seq(v)
```

```
##  [1]  1  2  3  4  5  6  7  8  9 10
```

```
length(v)
```

```
## [1] 10
```

```
sum(v)
```

```
## [1] 275
```

```
mean(v)
```

```
## [1] 27.5
```

A2 Create w that repeats the vector c(1,2,3) four times, and z that repeats each element of c(1, 2, 3) four times. Print both.

```
w <- rep(c(1,2,3), times = 4)
z <- rep(c(1, 2, 3), each = 4)
w
```

```
##  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

```
z
```

```
##  [1] 1 1 1 1 2 2 2 2 3 3 3 3
```

A3 Compute and print: the remainder of 17 divided by 5, the integer division of 17 by 5, 2 to the power of 10, the square root of 144, pi rounded to 3 decimals.

```
remainder <- 17 %% 5
remainder
```

```
## [1] 2
```

```
division <- 17 /2
division
```

```
## [1] 8.5
```

```
exponent <- 2^10
exponent
```

```
## [1] 1024
```

```r
sqrt(144)
```

```
## [1] 12
```

```r
round(pi,3)
```

```
## [1] 3.142
```

A4. Given a <- c(2, 4, 6, 8) and b <- c(1, 3, 5, 7), print a+b, a * b, logical comparison a > b, and the dot product sum(a*b)

```r
a <- c(2, 4, 6, 8)
b <- c(1, 3, 5, 7)
a + b
```

```
## [1]  3  7 11 15
```

```r
a * b
```

```
## [1]  2 12 30 56
```

```r
a > b
```

```
## [1] TRUE TRUE TRUE TRUE
```

```r
sum(a*b)
```

```
## [1] 100
```

A5 Use paste() to produce “Student-1” “Student-2” “Student-3”. Then identify the datatype of 42, 42L, “42”, 41i, and TRUE. What are its output?

```r
paste("Student", 1:3, sep = "-")
```

```
## [1] "Student-1" "Student-2" "Student-3"
```

```r
class(42)
```

```
## [1] "numeric"
```

```r
class(42L)
```

```
## [1] "integer"
```

```r
class("42")
```

```
## [1] "character"
```

```r
class(41i)
```

```
## [1] "complex"
```

```r
class(TRUE)
```

```
## [1] "logical"
```

Part B : Data Frames and Conditional Statements

B1

```r
students <- data.frame(
  id = 1:8,
  name = c("Alice","Bruno","Chen","Dana","Elif","Farid","Grace","Hiro"),
  sex = c("F","M","M","F","F","M","F","M"),
  section = c("A","B","A","C","B","C","A","B"),
```

```
  midterm = c(78,85,92,67,74,88,95,59),
  final = c(82,79,96,71,80,91,93,65),
  attendance = c(95,88,100,72,90,85,98,60),
  stringsAsFactors = FALSE
)
```

B2.

```
str(students)
```

```
## 'data.frame':    8 obs. of  7 variables:
##  $ id        : int  1 2 3 4 5 6 7 8
##  $ name      : chr  "Alice" "Bruno" "Chen" "Dana" ...
##  $ sex       : chr  "F" "M" "M" "F" ...
##  $ section   : chr  "A" "B" "A" "C" ...
##  $ midterm   : num  78 85 92 67 74 88 95 59
##  $ final     : num  82 79 96 71 80 91 93 65
##  $ attendance: num  95 88 100 72 90 85 98 60
```

```
dim(students)
```

```
## [1] 8 7
```

```
names(students)
```

```
## [1] "id"         "name"       "sex"        "section"    "midterm"
## [6] "final"      "attendance"
```

```
summary(students)
```

```
##        id              name               sex             section     midterm
##  Min.   :1.00   Length   :8   Length   :8   Length   :8   Min.   :59.00
##  1st Qu.:2.75   N.unique :8   N.unique :2   N.unique :3   1st Qu.:72.25
##  Median :4.50   N.blank  :0   N.blank  :0   N.blank  :0   Median :81.50
##  Mean   :4.50   Min.nchar:4   Min.nchar:1   Min.nchar:1   Mean   :79.75
##  3rd Qu.:6.25   Max.nchar:5   Max.nchar:1   Max.nchar:1   3rd Qu.:89.00
##  Max.   :8.00                                            Max.   :95.00
##      final         attendance
##  Min.   :65.00   Min.   : 60.00
##  1st Qu.:77.00   1st Qu.: 81.75
##  Median :81.00   Median : 89.00
##  Mean   :82.12   Mean   : 86.00
##  3rd Qu.:91.50   3rd Qu.: 95.75
##  Max.   :96.00   Max.   :100.00
```

B3.

```
students$average <- (students$midterm + students$final) /2
```

B4.

```
students$passed <- students$average >= 70 & students$attendance >= 75
```

B5.

```
students$standing <- ifelse(students$average >= 90, "Honors",
                     ifelse(students$average >= 75, "Good", "Needs Support"))
```

B6.

```r
classify <- function(score) {
  if (score >= 85) {
    "Distinction"
  } else if (score >= 70) {
    "Merit"
  } else if (score >= 50) {
    "Pass"
  } else {
    "Fail"
  }
}
result <- sapply(c(90, 73, 55, 40), classify)
print(result)   # "Distinction" "Merit" "Pass" "Fail"
```

```
## [1] "Distinction" "Merit"       "Pass"        "Fail"
```

Part C: rbind() and cbind()

C1.

```r
new_students <- data.frame(
  id = c(9, 10),
  name = c("Ines", "Jamal"),
  sex = c("F", "M"),
  section = c("C", "A"),
  midterm = c(81, 70),
  final = c(84, 68),
  attendance = c(92, 80),
  stringsAsFactors = FALSE
)

new_students$average <- (new_students$midterm + new_students$final) / 2

new_students$passed <- new_students$average >= 70 & new_students$attendance >= 75

new_students$standing <- ifelse(new_students$average >= 90, "Honors",ifelse(new_students$average >= 75,

students_all <- rbind(students, new_students)
```

C2.

```r
quiz <- c(8, 7, 10, 6, 7, 9, 10, 5, 8, 6)
cbind(students_all)
```

```
##    id  name sex section midterm final attendance average passed      standing
## 1   1 Alice   F       A      78    82         95    80.0   TRUE          Good
## 2   2 Bruno   M       B      85    79         88    82.0   TRUE          Good
## 3   3  Chen   M       A      92    96        100    94.0   TRUE        Honors
## 4   4  Dana   F       C      67    71         72    69.0  FALSE Needs Support
## 5   5  Elif   F       B      74    80         90    77.0   TRUE          Good
## 6   6 Farid   M       C      88    91         85    89.5   TRUE          Good
## 7   7 Grace   F       A      95    93         98    94.0   TRUE        Honors
## 8   8  Hiro   M       B      59    65         60    62.0  FALSE Needs Support
## 9   9  Ines   F       C      81    84         92    82.5   TRUE          Good
## 10 10 Jamal   M       A      70    68         80    69.0  FALSE Needs Support
```

```r
class(students_all)
```

```
## [1] "data.frame"
```

C3.

```r
m <- cbind(midterm = students_all$midterm, final = students_all$final)
m <- rbind(m, mean = colMeans(m))
print(m)
```

```
##      midterm final
##         78.0  82.0
##         85.0  79.0
##         92.0  96.0
##         67.0  71.0
##         74.0  80.0
##         88.0  91.0
##         95.0  93.0
##         59.0  65.0
##         81.0  84.0
##         70.0  68.0
## mean    78.9  80.9
```

C4.

```r
# rbind() on the data frames requires matching column names. If they differ,it stops with the error "nu
tryCatch(rbind(students, bad), error = function(e){message("rbind failed: ", conditionMessage(e)); NULL
```

```
## rbind failed: object 'bad' not found
```

```
## NULL
```

Part D: Subsetting

D1.

```r
students_all$name
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```r
students_all[["name"]]
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

```r
students_all[, "name"]
```

```
##  [1] "Alice" "Bruno" "Chen"  "Dana"  "Elif"  "Farid" "Grace" "Hiro"  "Ines"
## [10] "Jamal"
```

D2.

```r
students_all[2:5, c("name", "midterm", "final")]
```

```
##    name midterm final
## 2 Bruno      85    79
## 3  Chen      92    96
## 4  Dana      67    71
## 5  Elif      74    80
```

D3.

```
students_all[students_all$attendance >90, ]
```

```
##   id  name sex section midterm final attendance average passed standing
## 1  1 Alice   F       A      78    82         95    80.0   TRUE     Good
## 3  3  Chen   M       A      92    96        100    94.0   TRUE   Honors
## 7  7 Grace   F       A      95    93         98    94.0   TRUE   Honors
## 9  9  Ines   F       C      81    84         92    82.5   TRUE     Good
```

D4.

```
subset(students_all, sex == "F" & section %in% c("A", "C") & final >= 75, select = c(name, section, fina
```

```
##    name section final
## 1 Alice       A    82
## 7 Grace       A    93
## 9  Ines       C    84
```

D5.

```
students_all$name[which.max(students_all$average)]
```

```
## [1] "Chen"
```

```
students_all$name[which.min(students_all$average)]
```

```
## [1] "Hiro"
```

D6.

```
sorted <- students_all[order(students_all$average, decreasing = TRUE), ]
head(sorted, 3)
```

```
##   id  name sex section midterm final attendance average passed standing
## 3  3  Chen   M       A      92    96        100    94.0   TRUE   Honors
## 7  7 Grace   F       A      95    93         98    94.0   TRUE   Honors
## 6  6 Farid   M       C      88    91         85    89.5   TRUE     Good
```

Part E : Basic Factors

E1

```
students_all$section <- factor(students_all$section)
students_all$sex <- factor(students_all$sex)
levels(students_all$section)
```

```
## [1] "A" "B" "C"
```

```
levels(students_all$sex)
```

```
## [1] "F" "M"
```

E2

```
table(students_all$sex)
```

```
##
## F M
## 5 5
```

```
table(students_all$section)
```

```
##
## A B C
## 4 3 3
```

```
table(students_all$sex, students_all$section)
```

```
##
##     A B C
##   F 2 1 2
##   M 2 2 1
```

E3

```
as.integer(students_all$sex)
```

```
##  [1] 1 2 2 1 1 2 1 2 1 2
```

```
# Each integer is the index of the factor level: 1 = Female, 2 = Male categories. Factors store values
```

E4

```
students_all$sex_label <- factor(students_all$sex, levels = c("F", "M"),labels = c("Female", "Male"))

levels(students_all$sex_label)
```

```
## [1] "Female" "Male"
```

E5

```
summary(students_all$sex_label)
```

```
## Female   Male
##      5      5
```

```
mean(students_all$average[students_all$sex_label == "Female"])
```

```
## [1] 80.5
```

```
mean(students_all$average[students_all$sex_label == "Male"])
```

```
## [1] 79.3
```